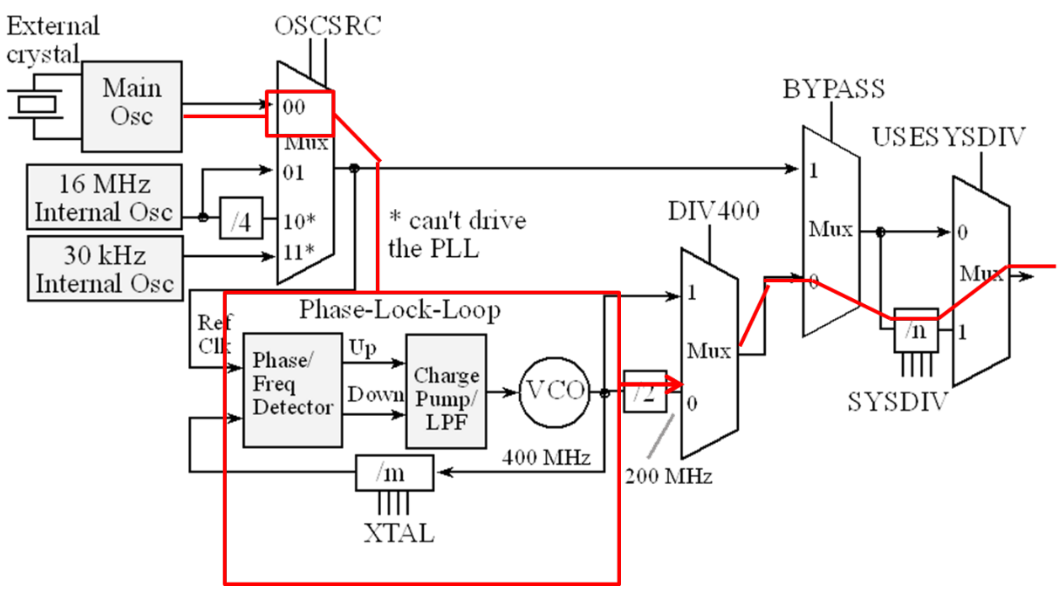
Name: ____________________

In-class Quiz
#09: System Clock and SysTick Timer

Note: All the codes are assumed to run on the TM4C123G microcontroller (TI LaunchPad).

1. The clock system of the TM4C123G has been configured in accordance with the following picture, where the line and box demonstrate the flow of the system clock. If the crystal frequency value of the Main OSC is 16 MHz, and the SYSDIV is configured to have a divisor of 10 (the frequency of the output clock is 1/10 of the frequency of the input clock). Please predict what the frequency of the system clock should be.



Answer: ____________________ MHz

2. If the crystal frequency is 10 MHz and the XTAL bits in the RCC register have been taken care of. What the frequency of the system clock should be. (All the other conditions are the same as in problem 1).

   Answer: ____________________ MHz

3. If the SysTick timer reload registers is loaded with the value 0x100, what is the counter value one clock cycle after reaching 0?

Answer: 0x____________________

4. As you already know it takes 2 cycles for the microcontroller to execute an `LDR` instruction. If in a specified LauchPad, it takes 100 ns (100×10^{-9} second) to execute an `LDR` instruction, what is the frequency of the system clock.

   Answer: ____________________

5. If the value in the SysTick Timer reload register is 4, please complete the following sequence to show the values in the SysTick Timer current value register along system clock cycles.

Clock	…	*n*	*n+1*	*n+2*	*n+3*	*n+4*	*n+5*	*n+6*	*n+7*	*n+8*	*n+9*	…
Values	…	2	1									…

6. Please choose the correct description about how to clear the count flag after it goes high. Circle your answer.

   a. The count flag will automatically go low after reading from the SysTick Timer Control register.
   b. The count flag will be low after you write any number to the SysTick Current register.

7. Check the following code, if the immediate value in the blank is `#0x480000`, what will be the number in the SysTick timer reload register? If the number in the blank is `#0x12340000`, what will be the number in the SysTick timer reload register?

```
      MOV  R0, #____________
SysTick_Wait
      LDR  R1, =NVIC_ST_RELOAD_R
      STR  R0, [R1]
```

   Answer: 0x____________________ if the immediate value is `0x480000`

   0x____________________ if the immediate value is `0x12340000`

8. What address would you read from if you want to know if the counter has ever hit zero.

   Answer: 0x____________________

9. Suppose the system clock is 16 MHz, the SysTick Timer and the PORTF have been configured and are ready to use. Please design a program to toggle the on/off status of the RED LED every 10 seconds. Write down your design and code in the following. You don't need to consider the few extra cycles due to checking the status of the timer and branching. The description of the design could be either of explanatory description, flow chart, pseudocode or structured English. It doesn't need to be accurate, just let people know your basic idea.
(Hint: Please check how long time the SysTick Timer can count.)